

Concepts of stripping ratios: Types of stripping ratios and their significance.

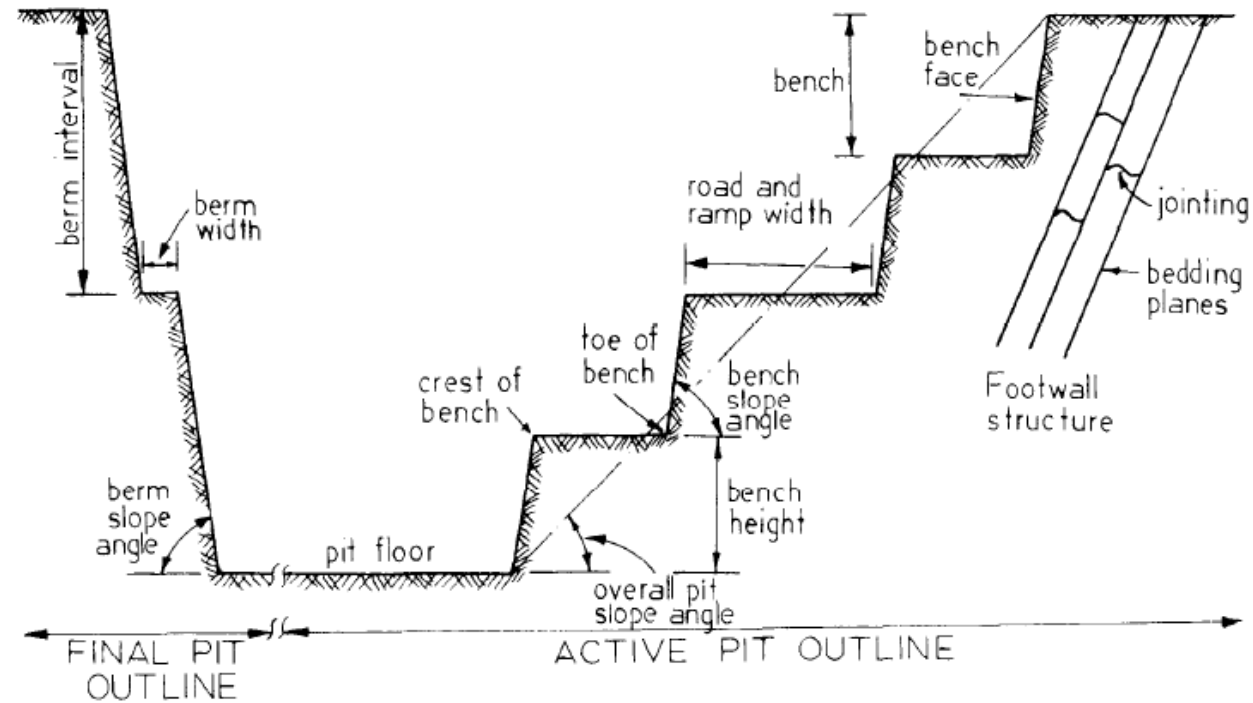
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Introduction

► Bench Geometry



It should be kept vertical but in practice it is difficult to maintain. Also it depends upon type of strata. Usually in practice it is kept to be 60°–80° to the horizontal for the working or active benches; and 45°–60° for non-working benches.

$$\text{Minimum Bench Width} = \text{Working Berm Width} + \text{Non-Working Berm Width} \quad (17.2a)$$

$$\text{Working berm width} = 3 \text{ times the width of the truck/dumper to be operated on the bench.} \quad (17.2b)$$

$$\text{Non-working berm width} = 3 \text{ m} \quad (17.2c)$$

$$\text{Thus, Bench Width} = 3 \times \text{Truck Width (Or width of largest equipment operating)} + 3 \text{ m} \quad (17.2d)$$

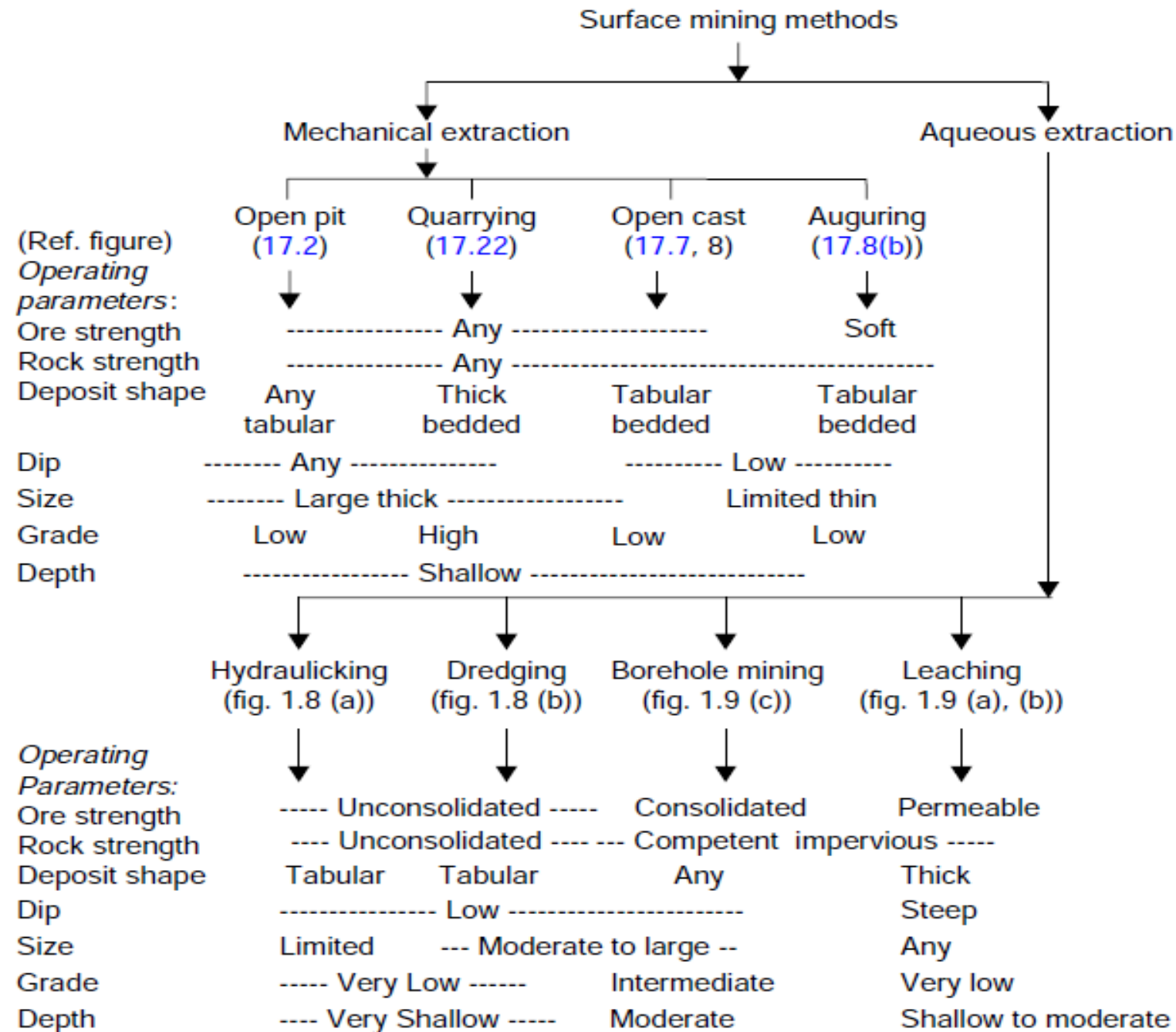
The Safety Berm is left when the bench reaches its ‘ultimate end’.

$$\text{Safety Berm} = 0.2 \times \text{Berm Interval (i.e. bench height)}^* \quad (17.3a)$$

$$= (1/3) \times \text{Berm Interval (i.e. bench height)}^{**} \quad (17.3b)$$

(* – Minimum Safety berm width as per Russian Safety Regulation)²⁴

(** – Minimum Safety berm width as per Hustrulid, Kuchta, 1998)¹⁴



17.2.2.1 Computation of overall pit slope angle

Figure¹⁴ 17.3 illustrates the geometry of an open pit. Geometrically overall pit slope angle can be computed as follows:¹⁴

$$OVERALL\ PITSLOPE\ (\phi) = \tan^{-1} \frac{N_B \times B_H}{\{(N_B - 1)B_W\} + \frac{N_B \times B_H}{\tan(B_A)} + (R_W)} \quad (17.4)$$

Whereas: ϕ – overall pit slope angle; degrees

N_B – number of benches

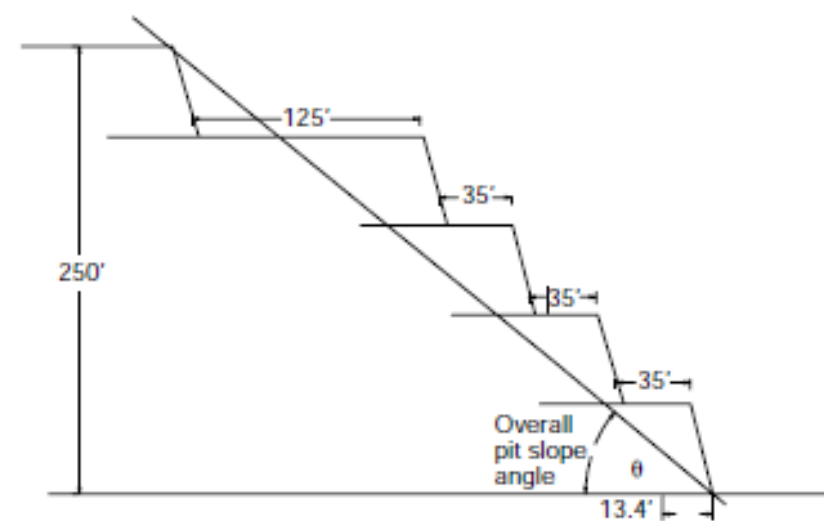
B_H – bench height in meters;

B_W – bench width in meters;

B_A – bench angle in degrees;

R_W – ramp width in meters (if intersected).

- Given data:
- (i) $N_B = 5$; $B_H = 16$ m; $B_W = 12$ m; $B_A = 75^\circ$; $R_W = 0$ m; (answer: 49°)
 - (ii) $N_B = 8$; $B_H = 10$ m; $B_W = 12$ m; $B_A = 75^\circ$; $R_W = 0$ m; (answer: 37.2°)
 - (iii) $N_B = 5$; $B_H = 16$ m; $B_W = 12$ m; $B_A = 75^\circ$; $R_W = 30$ m; (answer: 38.8°)



(a) Working pit slope angle is a function of number of benches and bench height

Types of stripping ratios and their significance.

- ▶ In open pit mines in order to decide the depth of the pit, it is essential to carry out the detailed calculations as how much waste rock will be required to remove to strip the ore body? The ratio between the amount of waste rock to be removed to mine out a unit of ore is called stripping ratio. Since it is ratio, and therefore, it should be dimensionless.

$$S.R = \text{Total waste (m}^3\text{) of / Total ore (tons.)}$$

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- ▶ To determine maximum depth based on the profitability of the operation, it is essential to know about the overall costs and revenues that will be received by selling the ore and its by-products, if any.
- ▶ In other words what will be cost of removing waste rock that is enclosing the ore body between surface datum and a particular depth, and also mining or exploiting the ore body itself that lies within this envelope.

$$SR_{max} = \frac{(Price\ or\ revenues\ received\ per\ ton\ of\ ore - cost\ of\ mining\ per\ ton.\ of\ ore)}{(Waste\ removing\ Cost/t)}$$

Stripping ratio is defined as the ratio of the Overburden/Cover Rock / Waste Rock to the ore body. The various stripping ratios that are used in surface mining are the following:

1. Volumetric / geometric stripping ratio
2. Break even stripping ration
3. Limiting stripping ratio
4. Instantaneous stripping ratio
5. Cumulative/ average stripping ratio
6. Overall Stripping ratio
7. Incremental stripping ratio

The *cut-off stripping ratio* is the one for which the costs of mining the ore and waste are matched by the revenue from that block of ore.

The ratio of the total volume of waste to the ore volume is defined as the *overall stripping ratio*:

$$R = \frac{\text{volume of waste removed to depth } d}{\text{volume of ore recovered to depth } d}$$

The *Instantaneous Stripping Ratio (ISR)* is the stripping ratio for a given *push back*, where a tiny slice of material, i.e., ore and/or waste, is removed from a pit wall.

Geometric / Volumetric stripping ratio (VSR):

$$\text{VSR} = \frac{\text{Volume of the overburden / waste rock, m}^3}{\text{Volume of ore, m}^3}$$

In case of horizontal or flat bedded deposits VSR becomes same as thickness ratio (TR).

$$\text{VSR} = \text{TR} = \frac{\text{Thickness of overburden, m}}{\text{Thickness of Ore bed, m}}$$

In case of simultaneous back filling thickness ratio helps in knowing whether the backfilled surface will be lower than, equal to or above the pre-mined land surface level.

Break even stripping ratio (BESR)

In case of coal deposits,

$$\text{BESR} = \frac{\text{Profit earned from coal mined in Rs/ te}}{\text{Cost of handling overburden in Rs / m}^3}$$

In case of other deposits,

$$\text{BESR} = \frac{\text{Profit earned from ore mined in Rs/ te}}{\text{Cost of handling waste rock in Rs / te}}$$

BESR helps in determining the pit limit.

Maximum allowable (break-even)stripping ratio

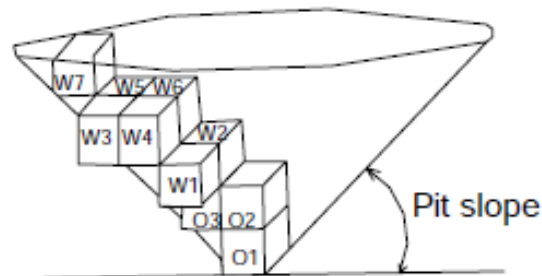
The maximum allowable stripping ratio ($SR_{\max}$) is significant because it is purely economic ratio and determines the pit limit at current economic scenario. Cost is applied to the physical waste and ore tonnages and that determines the profitability.

Each scheme has its own merits and demerits.

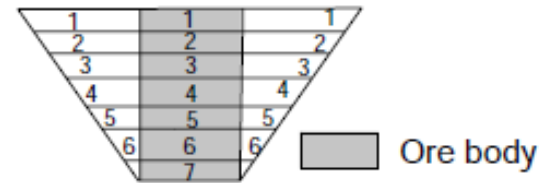
Declining SR method (fig.(d)) put up heavy burden on production of waste in the beginning that may adversely affect the cash-flow in the earlier years.

Increasing SR method (fig.(e)) produces large cash-flow in the initial period and covers the risk, and that's why, it is preferred in most of the cases. sometimes it is impractical to operate large number of benches and faces, but keeping them open (active) gives an opportunity to blend the ore.

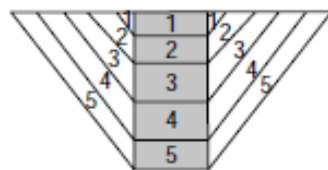
The constant SR method (fig.(f)) is the compromise of the extreme conditions that are associated with the other two methods. In practice a method, which can generate high cash flow and require building up of resources (man and equipment) gradually, and reduction of resources gradually at the end of pit-life should be preferred.



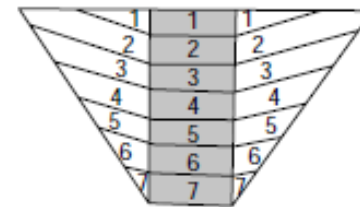
(c) Open pit coning at break-even scenario.
O is a ore block. W is a waste block.



(d) Declining stripping ratio



(e) Increasing stripping ratio



(f) Constant stripping ratio

Importance

- ▶ The stripping ratio method of calculating the profitability of the mining procedure is commonly used in the ore or coal mining process. In relation to its application in the mining of ore, the utility of **the stripping ratio is the determination of the approximate ratio of waste material that must be moved in order to reach a certain ratio of ore.** As can be seen, when the ratio of unwanted material that must be moved in order to reach the desired ore far outweighs any ore that can potentially be extracted, then it can be concluded that the input in terms of labor, time and material resources will far outweigh any benefits that can be gained from mining.
- ▶ The stripping ratio is important **due to the fact that it helps a mining company assess a mining site in order to determine the profitability before committing any resources to the mining process.** A way of describing the concept of stripping ratio is by looking at it in terms of the **level at which the mining process will cease to be profitable.** For example, if the mining company must extract one part of ore to three and a half parts of waste in order to break even, it stands to reason that the company will seek to look for mining sites where it can extract at least one part of ore to two and a half parts of waste in order for the mining venture to be profitable. The waste in question includes debris, sand and different sizes of rocks, which will be moved from their location in order to reach the ore beneath.
- ▶ The extraction of the ore through the application of the **stripping ratio also takes into consideration the density of the material that must be moved in order to reach the ore and other factors that include the heaviness of the material.** For example, lightweight top material may be easier to remove from its location, while denser and heavier material might make it more difficult, contributing to an increase of the stripping ratio in favor of the mining company. Another consideration that affects the ratio is the presence of contaminants that might lower the quality of the material extracted from the site, meaning that more of the material has to be extracted in order to meet up with the profit target.

Thank you